Questions and Answers

1. Which courier handled the most parcels?

Boult – 1924 parcels handled

2. What is the overall average shipping time (days) for delivered parcels?

2.60 days

3. Which region has the longest average shipping time?

North – 2.62 days on average

4. Which product category generated the highest total order value?

Electronics – total value = 154 283.41

5. What is the refund rate (number of orders refunded divided by all orders) for each product category? Which is worst?

Category	Refund Rate
Clothing	0.0906
Toys	0.0889
Electronics	0.0814
Sports	0.0731
Home	0.0707

Clothing has the worst refund rate

6. Is the gap between best and worst category refund rates statistically significant? Briefly explain your approach and test.

The data is comparing proportions from independent groups; in this case we are comparing two independent groups. This leads me to determine a two-proportion z-test to be the best statistical test to determine if the refund rate indexes for Clothing and Home differ significantly.

The z-test returned a z-statistic of -1.756 and a p-value of 0.0792.

This P-value is greater than the accepted threshold of 0.05, leading me to accept my null hypothesis and declare there is no significant statistical difference in the refund rates between the clothing and home categories

7. Which courier has the lowest percentage of their parcels that were returned AND refunded?

Arrow has the lowest % of parcels that were returned and refunded at 2.08%

8. Calculate net revenue per parcel for each combination of courier and product category (eg. Arrow–Home, Cyclone–Clothing). Which combination yields the highest net revenue per parcel? State any assumptions

Assumptions:

- Each shipment_id relates to an individual parcel, one order_id may have multiple parcels attached as a result, multiple items may be shipped in the same parcel.
- Customer cancellations will be eaten by the firm and not courier, so should not be included in the courier's net revenue per parcel.
- Incorrect items are treated as errors by the online marketplace and not fulfilment errors by the courier.
- All other refund reasons e.g damaged/late are to be included in net revenue per parcel.
- If a shipment is lost, the total order value is factored as a cost.

The highest net revenue per parcel is achieved by Cyclone delivering Electronics with a net revenue per parcel of 114.358